220110: sandstone, Wendy 1

(Tumblagooda Sandstone?, northern Perth Basin)

Location and sampling

Geraldton (SH 50-1), Mungo (1941)
MGA Zone 50, 305515E 6868070N

Sampled on 24 March 2017

This sample was collected from the 783.5 – 784.5 m depth interval of drillcore Wendy 1, a stratigraphic well drilled in 2004 by Victoria International Petroleum NL (Victoria Petroleum, 2004). The drillhole is located on the eastern side of Dartmoor Road, about 30 km northeast of Mount Erin, 13.4 km west of Wandana Hill, and 3.2 km north of the Yuna–Tenindewa Road.

Geological context

The unit sampled is tentatively assigned to the 419–393 Ma Tumblagooda Sandstone (Hocking, 1991; McNamara, 2014; Hocking and Allen, 2021), or may be a separate unit coeval with or slightly younger than the Tumblagooda Sandstone. The siliciclastic interval from 112.0 to 1210.5 m in Wendy 1 has previously been interpreted as Tumblagooda Sandstone (Victoria Petroleum, 2004; Retallack, 2009). Kettanah et al. (2015) tentatively interpreted a stratigraphic break at 642 m (or possibly as deep as 793 m), with only the interval below this break (to a depth of 1210 m) interpreted as the Tumblagooda Sandstone. However, hyperspectral data indicates the presence of carbonate at depths of 780–840 m, which is inconsistent with detailed sedimentological studies of the Tumblagooda Sandstone (Allen and Trinajstic, 2017). A maximum age for deposition of the Tumblagooda Sandstone is provided by a 434 ± 16 Ma Pb-isotope age for Pb mineralization in stratigraphically underlying metamorphic rocks of the Northampton Inlier (Richards et al., 1985). A laser ablation U–Pb detrital zircon study of 16 Tumblagooda Sandstone samples yielded a youngest zircon at 466 ± 8 Ma (1σ) and a conservative maximum depositional age of 481 ± 4 Ma (GSWA 214841, Markwitz et al., 2017). Based on comparable trace fossil assemblages and plant remains (H Allen, 2021, written comm., 20 October) the Tumblagooda Sandstone may be latest Silurian to Early Devonian in age. Zircons from a sandstone sample taken from about 323 m depth in Wendy 1 yielded an age of 497 ± 4 Ma for the youngest zircon and a conservative maximum depositional age of 501 ± 8 Ma (GSWA 220109, Wingate et al., 2021).

Petrographic description

The sample is a pale yellow–grey, quartz-rich sandstone (Fig. 1). A detailed petrographic description is not available for this sample. However, petrography of other Tumblagooda Sandstone samples, including some from Wendy 1, is reported by Kettanah et al. (2015, and references within).

Zircon morphology

Zircons isolated from this sample are mainly colourless (a few are pale yellow or pale purple), anhedral to euhedral, and strongly rounded. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 6:1. Most crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, exhibit concentric zoning truncated at grain edges, features consistent with abrasion during sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 12 June 2018, using LA-ICP-MS instruments at the GeoHistory Facility in the John de Laeter Centre, Curtin University. Separated zircons mounted in epoxy disks were ablated using a Resonetics RESOlution M-50A-LR sampling system, incorporating a 193 nm Compex 102 excimer laser, and ion signal intensities were measured using an Agilent 7700s quadrupole ICP-MS. Samples were ablated for 35 s at a 5 Hz repetition rate, using a 38 µm beam diameter and laser energy of 2.5 J/cm^2. Seventeen analyses of the 91 500 zircon standard obtained during the session indicated a concordia age of 1062.7 ± 2.1 Ma (MSWD = 0.84), in agreement with the accepted value of 1062.4 Ma. Secondary zircon standards GJ1, OGC1 and Plešovice also yielded mean dates in agreement with their accepted values. Additional analytical details are provided in Fielding et al. (2021). All relevant uncertainties have been propagated into the ratios and dates reported in Table 1. Dates for analyses for which $^{238}U/^{206}Pb$ ratios indicate dates <1000 Ma are based on 207-corrected $^{238}U/^{206}Pb^*$ ratios; those >1000 Ma are based on uncorrected $^{207}Pb/^{206}Pb$ ratios.

Results

A total of 150 analyses were obtained from 150 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3, 4) and a probability density plot (Fig. 5).

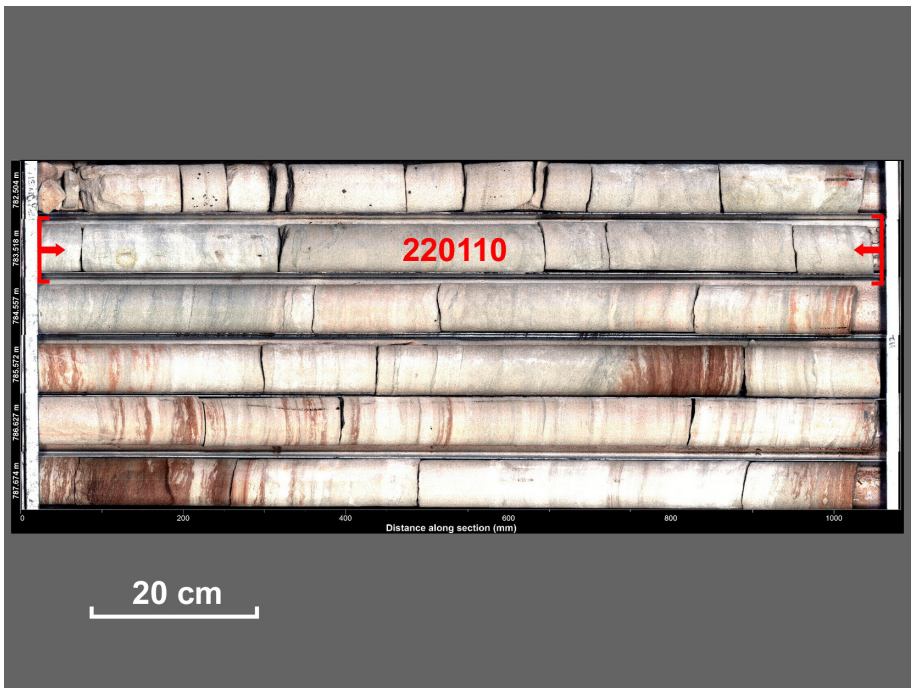

Figure 1. Drillcore (HyLogger-3) image of sample 220110: sandstone, Wendy 1. The sample interval is highlighted in red

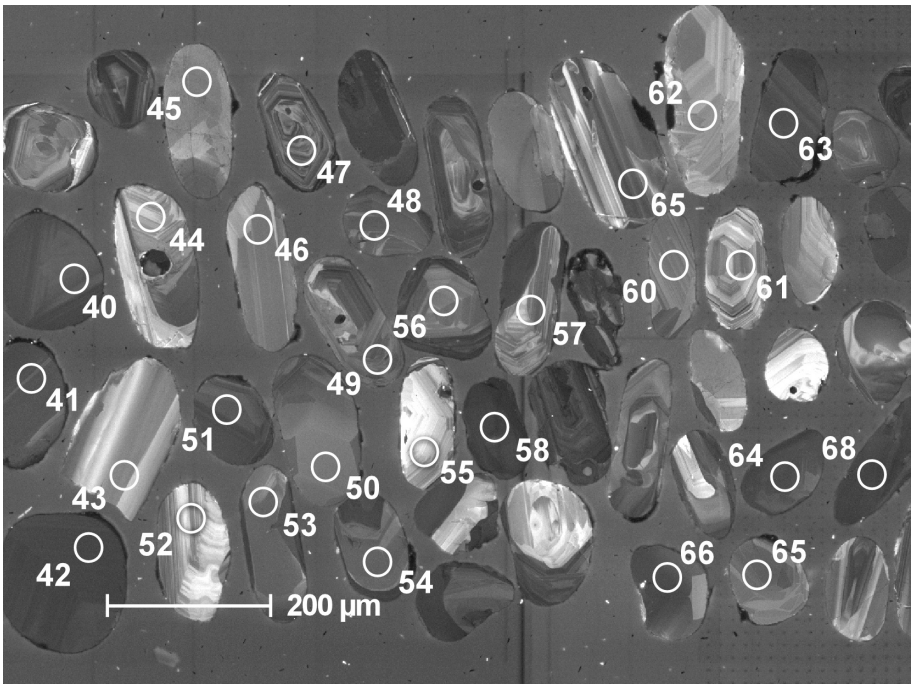

Figure 2. Cathodoluminescence image of representative zircons from sample 220110: sandstone, Wendy 1. Numbered circles indicate the approximate locations of analysis sites

Table 1. LA-ICP-MS analytical results for zircons from sample 220110: sandstone, Wendy 1

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		Disc. (%)
Y	17	17	75	125	1.67	12.282	0.072	0.06080	0.00120	505	3	632	43	20.2
S	3	3	100	229	2.29	12.315	0.058	0.05680	0.00095	503	2	484	37	-4.0
S	120	120	177	363	2.05	11.902	0.052	0.05750	0.00075	520	2	511	29	-1.8
S	147	147	102	218	2.15	11.805	0.065	0.06210	0.00120	524	3	678	41	22.6
S	150	150	188	287	1.53	11.830	0.053	0.05870	0.00080	523	2	556	30	5.9
S	99	99	95	117	1.24	11.791	0.067	0.05770	0.00095	525	3	518	36	-1.2
S	36	36	193	239	1.23	11.699	0.053	0.05780	0.00070	529	2	522	27	-1.3
S	133	133	149	451	3.03	11.689	0.050	0.05720	0.00075	529	2	499	29	-6.0
S	57	57	123	264	2.15	11.636	0.062	0.05940	0.00085	531	3	582	31	8.6
S	52	52	62	119	1.92	11.628	0.081	0.05890	0.00120	532	4	563	44	5.6
S	141	141	74	114	1.54	11.628	0.074	0.05570	0.00105	532	3	440	42	-20.8
S	122	122	173	540	3.12	11.501	0.062	0.05820	0.00075	537	3	537	28	0.0
S	132	132	197	223	1.13	11.444	0.054	0.06080	0.00080	540	2	632	28	14.6
S	91	91	85	248	2.91	11.416	0.072	0.05940	0.00100	541	3	582	37	7.0
S	112	112	134	138	1.03	11.370	0.059	0.05740	0.00080	543	3	507	31	-7.2
S	149	149	212	198	0.93	11.228	0.061	0.06050	0.00065	550	3	622	23	11.5
S	31	31	310	403	1.30	10.955	0.046	0.05776	0.00055	563	2	521	21	-8.1
S	79	79	183	184	1.01	10.653	0.045	0.06110	0.00070	578	2	643	25	10.0
S	21	21	102	87	0.86	10.618	0.050	0.06010	0.00090	580	3	607	32	4.4
S	35	35	107	101	0.95	10.465	0.051	0.05920	0.00085	588	3	574	31	-2.4
S	37	37	403	172	0.43	10.221	0.036	0.05940	0.00055	602	2	582	20	-3.4
S	43	43	78	205	2.63	10.079	0.056	0.06130	0.00105	610	3	650	37	6.1
S	5	5	83	28	0.34	10.030	0.055	0.06130	0.00095	613	3	650	33	5.7
S	92	92	116	24	0.21	9.990	0.060	0.06140	0.00080	615	4	653	28	5.9
S	32	32	200	53	0.26	9.975	0.042	0.06140	0.00070	616	2	653	24	5.7
S	28	28	185	165	0.89	9.946	0.040	0.05900	0.00065	618	2	567	24	-8.9
S	64	64	319	164	0.51	9.770	0.043	0.06088	0.00055	628	3	635	19	1.1
S	45	45	115	65	0.57	9.157	0.046	0.06370	0.00080	668	3	732	27	8.7
S	81	81	95	58	0.60	8.818	0.051	0.06230	0.00085	692	4	684	29	-1.2
S	10	10	177	134	0.76	8.636	0.048	0.06584	0.00065	706	4	801	21	11.8
S	41	41	266	207	0.78	8.648	0.036	0.06296	0.00060	705	3	707	20	0.2
S	15	15	224	149	0.66	7.988	0.032	0.06450	0.00070	760	3	758	23	-0.3
S	137	137	136	65	0.47	7.799	0.036	0.06640	0.00070	778	3	819	22	5.0
S	19	19	480	278	0.58	7.764	0.039	0.06630	0.00047	781	4	816	15	4.3
S	72	72	322	216	0.67	7.720	0.027	0.06522	0.00055	785	3	781	18	-0.5
S	131	131	132	82	0.62	7.686	0.035	0.06440	0.00075	788	3	755	25	-4.5
S	101	101	113	121	1.07	7.632	0.035	0.06560	0.00085	794	3	794	27	0.0

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	Disc. (%)
S	127	127	222	132	0.59	7.467	0.031	0.06483	0.00060	810	3	769	19	-5.4
S	82	82	434	470	1.08	7.404	0.027	0.06615	0.00055	817	3	811	17	-0.7
S	73	73	142	113	0.79	7.132	0.031	0.06680	0.00070	846	3	832	22	-1.7
S	78	78	328	166	0.50	6.744	0.025	0.06777	0.00055	891	3	862	17	-3.5
S	39	39	300	227	0.76	6.676	0.027	0.06806	0.00050	900	3	870	15	-3.4
S	11	11	170	82	0.48	6.646	0.027	0.06890	0.00065	904	3	896	19	-0.9
S	27	27	53	60	1.12	6.588	0.039	0.06880	0.00100	911	5	893	30	-2.0
S	76	76	304	248	0.82	6.569	0.024	0.06942	0.00055	914	3	911	16	-0.2
S	14	14	78	24	0.31	6.553	0.030	0.07030	0.00080	916	4	937	23	2.3
S	46	46	126	195	1.55	6.502	0.030	0.07150	0.00080	922	4	972	23	5.1
S	77	77	64	38	0.60	6.485	0.042	0.07040	0.00100	924	6	940	29	1.7
S	62	62	85	43	0.50	6.460	0.031	0.07090	0.00080	928	4	955	23	2.8
S	142	142	116	132	1.14	6.435	0.029	0.07030	0.00075	931	4	937	22	0.6
S	63	63	314	141	0.45	6.431	0.025	0.07019	0.00060	932	3	934	18	0.2
S	60	60	131	109	0.83	6.406	0.029	0.07180	0.00070	935	4	980	20	4.6
S	100	100	115	38	0.33	6.418	0.029	0.06980	0.00075	933	4	922	22	-1.2
S	18	18	299	155	0.52	6.415	0.025	0.07011	0.00060	934	3	932	18	-0.2
S	42	42	498	201	0.40	6.394	0.025	0.06970	0.00050	937	3	920	15	-1.9
S	49	49	176	210	1.19	6.386	0.029	0.07050	0.00075	938	4	943	22	0.5
S	88	88	500	165	0.33	6.382	0.026	0.07059	0.00050	938	4	946	15	0.8
S	117	117	361	479	1.33	6.322	0.022	0.07063	0.00055	947	3	947	16	0.0
S	40	40	227	141	0.62	6.297	0.042	0.07148	0.00060	950	6	971	17	2.2
S	102	102	162	95	0.58	6.281	0.032	0.07200	0.00070	952	4	986	20	3.4
S	71	71	92	77	0.83	6.277	0.030	0.07210	0.00080	953	4	989	23	3.6
S	90	90	186	115	0.62	6.277	0.028	0.07137	0.00055	953	4	968	16	1.6
S	134	134	415	197	0.48	6.262	0.025	0.07036	0.00055	955	4	939	16	-1.7
S	22	22	874	79	0.09	6.258	0.025	0.07074	0.00048	956	4	950	14	-0.6
S	51	51	253	3	0.01	6.250	0.025	0.07148	0.00060	957	4	971	17	1.5
S	146	146	363	160	0.44	6.247	0.023	0.07091	0.00055	957	3	955	16	-0.3
S	111	111	471	165	0.35	6.245	0.025	0.07072	0.00050	957	4	949	14	-0.9
S	16	16	123	418	3.40	6.234	0.027	0.07180	0.00085	959	4	980	24	2.2
S	108	108	146	79	0.54	6.238	0.027	0.07117	0.00060	958	4	962	17	0.4
S	61	61	78	37	0.48	6.231	0.037	0.07050	0.00095	960	5	943	28	-1.8
S	20	20	204	255	1.25	6.219	0.027	0.07090	0.00060	961	4	955	17	-0.7
S	121	121	97	84	0.87	6.219	0.029	0.07080	0.00075	961	4	952	22	-1.0
S	67	67	267	163	0.61	6.207	0.029	0.07130	0.00060	963	4	966	17	0.3
S	130	130	158	90	0.57	6.188	0.027	0.07260	0.00070	966	4	1003	20	3.7
S	30	30	179	140	0.78	6.150	0.025	0.07115	0.00065	971	4	962	19	-1.0

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	Disc. (%)
S	7	7	90	20	0.22	6.109	0.063	0.07320	0.00080	977	9	1019	22	4.1
S	12	12	109	49	0.45	6.101	0.030	0.07190	0.00075	978	4	983	21	0.5
S	68	68	605	253	0.42	6.098	0.024	0.07190	0.00049	979	4	983	14	0.4
S	144	144	145	122	0.84	5.787	0.025	0.07181	0.00060	1028	4	981	17	-4.8
S	139	139	73	34	0.47	6.075	0.033	0.07240	0.00105	982	5	997	29	1.5
S	104	104	111	73	0.66	5.817	0.027	0.07190	0.00075	1023	4	983	21	-4.0
S	123	123	172	142	0.83	6.050	0.029	0.07200	0.00065	986	4	986	18	0.0
S	143	143	165	116	0.70	6.017	0.027	0.07220	0.00070	991	4	992	20	0.0
S	8	8	455	125	0.27	6.000	0.022	0.07166	0.00055	994	3	976	16	-1.8
S	128	128	227	508	2.24	5.959	0.025	0.07201	0.00055	1000	4	986	16	-1.4
S	138	138	167	224	1.34	5.917	0.028	0.07300	0.00070	1007	4	1014	19	0.7
S	135	135	221	220	0.99	5.797	0.024	0.07309	0.00065	1026	4	1016	18	-0.9
S	26	26	144	187	1.30	5.917	0.026	0.07340	0.00070	1007	4	1025	19	1.8
S	145	145	280	444	1.58	5.917	0.026	0.07369	0.00060	1007	4	1033	16	2.6
S	9	9	101	74	0.73	5.698	0.028	0.07380	0.00080	1042	5	1036	22	-0.6
S	2	2	238	107	0.45	5.528	0.021	0.07410	0.00065	1072	4	1044	18	-2.7
S	98	98	444	59	0.13	5.695	0.037	0.07413	0.00055	1043	6	1045	15	0.2
S	115	115	159	55	0.35	5.587	0.023	0.07450	0.00070	1062	4	1055	19	-0.6
S	87	87	174	153	0.87	5.482	0.023	0.07470	0.00070	1080	4	1060	19	-1.9
S	48	48	245	93	0.38	5.543	0.023	0.07497	0.00060	1069	4	1068	16	-0.1
S	86	86	153	142	0.93	5.602	0.024	0.07501	0.00065	1059	4	1069	17	0.9
S	66	66	142	143	1.01	5.382	0.022	0.07510	0.00065	1099	4	1071	17	-2.6
S	103	103	230	28	0.12	5.593	0.022	0.07518	0.00060	1060	4	1073	16	1.2
S	50	50	94	53	0.56	5.599	0.027	0.07520	0.00085	1059	5	1074	23	1.4
S	106	106	228	175	0.77	5.411	0.022	0.07521	0.00065	1093	4	1074	17	-1.8
S	114	114	228	68	0.30	5.540	0.021	0.07529	0.00060	1070	4	1076	16	0.6
S	113	113	139	48	0.34	5.574	0.026	0.07570	0.00070	1064	5	1087	19	2.2
S	126	126	106	70	0.66	5.615	0.024	0.07570	0.00065	1057	4	1087	17	2.8
S	44	44	73	109	1.50	5.485	0.026	0.07580	0.00090	1080	5	1090	24	0.9
S	124	124	91	51	0.56	5.420	0.026	0.07630	0.00080	1092	5	1103	21	1.0
S	118	118	112	180	1.60	5.391	0.028	0.07650	0.00080	1097	5	1108	21	1.0
S	25	25	30	7	0.24	5.498	0.042	0.07660	0.00145	1077	8	1111	38	3.0
S	125	125	165	98	0.60	5.510	0.024	0.07670	0.00070	1075	4	1113	18	3.4
S	53	53	165	121	0.73	5.181	0.024	0.07700	0.00075	1138	5	1121	19	-1.5
S	136	136	212	128	0.60	5.105	0.023	0.07720	0.00070	1153	5	1126	18	-2.4
S	59	59	183	418	2.29	5.071	0.021	0.07727	0.00065	1160	4	1128	17	-2.8
S	109	109	304	196	0.64	5.327	0.018	0.07780	0.00055	1109	4	1142	14	2.9
S	80	80	308	266	0.86	5.160	0.021	0.07815	0.00060	1142	4	1151	15	0.8

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$\frac{^{232}Th}{^{238}U}$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	Disc. (%)
S	119	119	59	23	0.39	5.187	0.031	0.07840	0.00105	1137	6	1157	27	1.8
S	83	83	155	103	0.66	4.943	0.022	0.07856	0.00065	1188	5	1161	16	-2.3
S	94	94	310	122	0.39	5.005	0.020	0.07861	0.00060	1174	4	1162	15	-1.0
S	4	4	81	183	2.26	5.023	0.023	0.07960	0.00085	1170	5	1187	21	1.4
S	34	34	401	127	0.32	4.904	0.019	0.07988	0.00060	1196	4	1194	15	-0.2
S	55	55	74	42	0.57	4.955	0.023	0.08020	0.00095	1185	5	1202	23	1.4
S	29	29	399	375	0.94	4.523	0.017	0.08315	0.00060	1288	4	1273	14	-1.2
S	56	56	89	77	0.86	4.539	0.021	0.08350	0.00080	1283	5	1281	19	-0.2
S	75	75	414	107	0.26	4.284	0.016	0.08706	0.00060	1352	4	1362	13	0.7
S	47	47	214	125	0.58	3.903	0.019	0.09120	0.00065	1470	6	1451	14	-1.4
S	13	13	241	220	0.91	4.112	0.019	0.09228	0.00070	1403	6	1473	14	4.7
S	89	89	139	89	0.64	3.138	0.012	0.10800	0.00085	1783	6	1766	14	-1.0
S	95	95	98	56	0.57	3.168	0.014	0.10820	0.00095	1769	7	1769	16	0.0
S	110	110	375	508	1.35	2.912	0.011	0.11576	0.00075	1903	6	1892	12	-0.6
S	33	33	349	142	0.41	2.907	0.010	0.11739	0.00075	1906	6	1917	11	0.6
S	65	65	205	78	0.38	2.563	0.011	0.12987	0.00090	2124	7	2096	12	-1.3
S	85	85	140	53	0.38	2.534	0.011	0.13172	0.00085	2144	8	2121	11	-1.1
S	96	96	215	59	0.28	2.259	0.010	0.14839	0.00095	2363	9	2327	11	-1.5
S	23	23	141	83	0.59	2.298	0.011	0.15420	0.00110	2329	9	2393	12	2.7
S	97	97	157	184	1.17	2.220	0.011	0.15425	0.00105	2397	10	2394	12	-0.1
S	107	107	216	309	1.43	2.135	0.009	0.16010	0.00105	2477	9	2457	11	-0.8
S	70	70	377	385	1.02	2.126	0.007	0.16077	0.00100	2485	7	2464	11	-0.9
S	69	69	160	151	0.94	2.131	0.010	0.16182	0.00105	2480	9	2475	11	-0.2
S	24	24	52	50	0.96	2.227	0.011	0.16230	0.00140	2391	10	2480	15	3.6
S	38	38	156	130	0.83	2.071	0.009	0.16690	0.00110	2540	9	2527	11	-0.5
S	148	148	113	109	0.97	2.017	0.011	0.16870	0.00115	2596	12	2545	11	-2.0
S	84	84	103	75	0.72	1.936	0.009	0.18010	0.00120	2685	10	2654	11	-1.2
S	6	6	380	115	0.30	2.013	0.007	0.18124	0.00115	2600	8	2664	11	2.4
S	105	105	231	185	0.80	1.920	0.007	0.18566	0.00120	2703	8	2704	11	0.0
S	129	129	267	174	0.65	1.781	0.007	0.20665	0.00130	2873	9	2879	10	0.2
D	140	140	141	113	0.80	9.398	0.093	0.07450	0.00325	652	6	1055	88	38.2
D	74	74	160	299	1.87	6.557	0.037	0.09570	0.00180	915	5	1542	35	40.7
D	93	93	54	23	0.44	5.750	0.031	0.07760	0.00100	1034	5	1137	26	9.1
D	1	1	206	225	1.09	3.925	0.015	0.09567	0.00075	1463	5	1541	15	5.1
D	116	116	252	186	0.74	2.593	0.013	0.15210	0.00105	2102	9	2370	12	11.3
D	54	54	200	139	0.70	2.438	0.012	0.15590	0.00105	2215	9	2412	11	8.1
D	58	58	214	151	0.70	1.982	0.008	0.16405	0.00105	2633	9	2498	11	-5.4

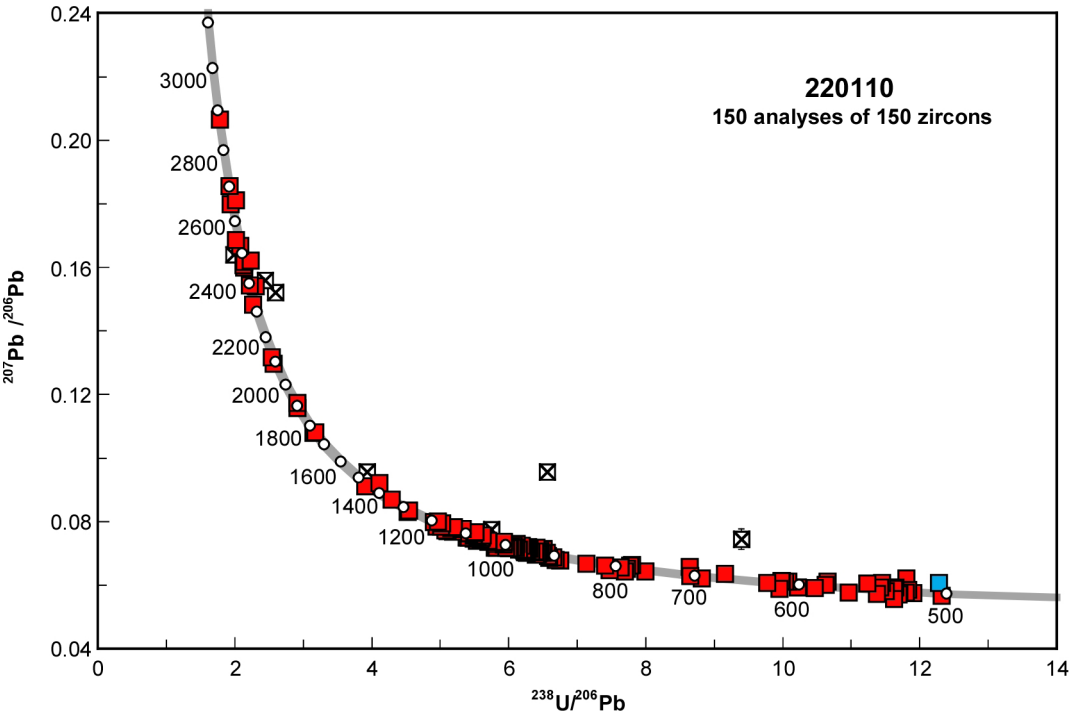

Figure 3. U–Pb analytical data, not corrected for common Pb, for sample 220110: sandstone, Wendy 1. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (date >1000 Ma with discordance >5% and date <1000 Ma with ƒ207 >1%)

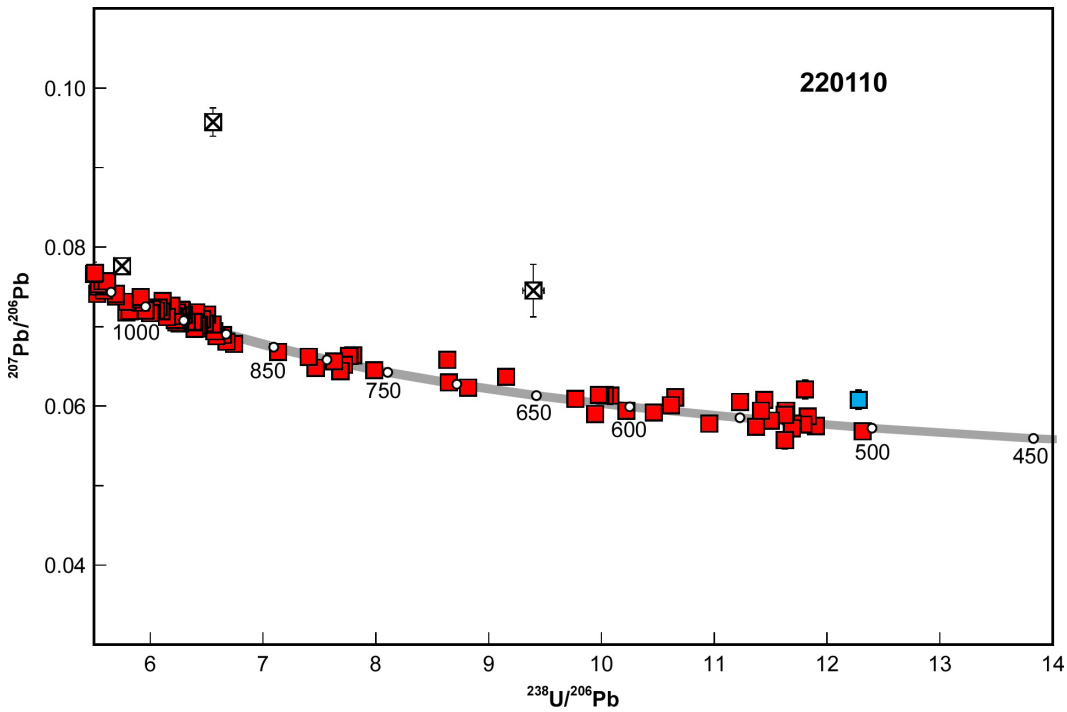

Figure 4. Expanded view of U–Pb analytical data, not corrected for common Pb, for sample 220110: sandstone, Wendy 1. Symbols as in Figure 3

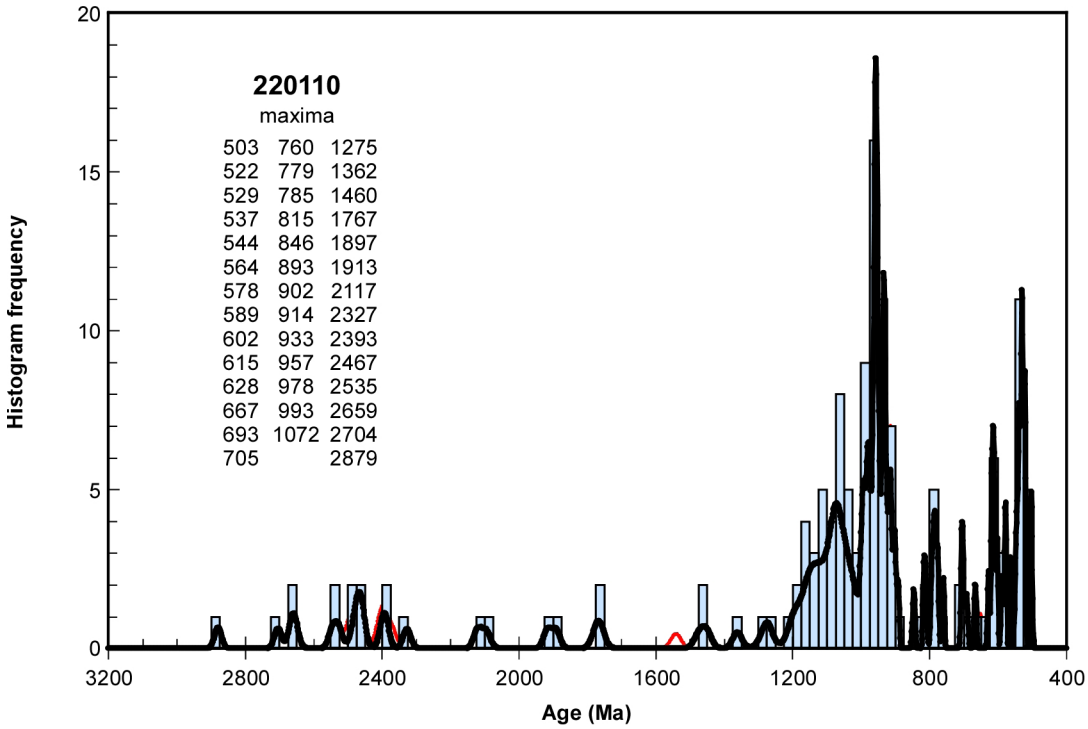

Figure 5. Probability density diagram and histogram for sample 220110: sandstone, Wendy 1. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) only include accepted data (143 analyses of 143 zircons). Thin curve includes all data (150 analyses of 150 zircons)

Interpretation

The analyses are concordant to strongly discordant (Figs 3, 4). Five analyses >1000 Ma are >5% discordant, and two analyses with U/Pb dates <1000 Ma have high common Pb (*f*207 >1%). The dates obtained from these seven analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 143 analyses can be divided into two groups, based on their $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a 207-corrected $^{238}U/^{206}Pb^*$ date of 502 ± 3 Ma (1σ).

Group S comprises 142 analyses (Table 1), which yield $^{207}Pb/^{206}Pb$ or $^{238}U/^{206}Pb^*$ dates of 2879–504 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 502 ± 3 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{238}U/^{206}Pb^*$ date of 524 ± 4 Ma (MSWD = 2.0) for the youngest coherent group of five analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 2467, 1072, 993, 978, 957, 933, 914, 902, 785, 615, 544, 537, 529, and 522 Ma, based on contributions from approximately 4, 18, 10, 8, 19, 10, 5, 3, 3, 5, 3, 5, 5, and 4 analyses, respectively (Fig. 5). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

References

Allen, HJ and Trinajstic, K 2017, An Early Devonian fish fauna from an unnamed sandstone in Wendy 1 drillhole, northern Perth Basin: Geological Survey of Western Australia, Paleontology Report 2017/1, 4p.

Fielding, IOH, Lu, Y and Wingate, MTD 2021, Introduction to geochronology information 2021: Geological Survey of Western Australia, 8p.

Hocking, RM 1991, The Silurian Tumblagooda Sandstone, Western Australia: Geological Survey of Western Australia, Report 27, 124p.

Hocking, RM and Allen, HJ (partial report), Tumblagooda Sandstone (D-_tu-st): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 07 October 2021, <www.dmp.wa.gov.au/ens>.

Kettanah, YA, Mory, AJ, Wach, GD and Wingate, MTD 2015, Provenance of the Ordovician – Lower Silurian Tumblagooda Sandstone, Western Australia: Australian Journal of Earth Sciences, v. 62, p. 817–830.

Markwitz, V, Kirkland, CL, Wyrwoll, KH, Hancock, EA, Evans, NJ and Lu Y 2017, Variations in zircon provenance constrain age and geometry of an Early Paleozoic rift in the Pinjarra Orogen, East Gondwana: Tectonics, v. 36, p. 2477–2496.

McNamara, KJ 2014, Early Paleozoic colonization of the land: evidence from the Tumblagooda Sandstone, Southern Carnarvon Basin, Western Australia: Royal Society of Western Australia Journal, v. 97(1), p. 111–132.

Retallack, GJ 2009, Cambrian, Ordovician and Silurian pedostratigraphy and global events in Australia: Australian Journal of Earth Sciences, v. 56, p. 571–586.

Richards, JR, Blockley, JG and de Laeter, JR 1985, Rb–Sr and Pb isotope data from the Northampton Block, Western Australia: Bulletin and Proceedings of the Australasian Institute of Mining and Metallurgy, v. 290, p. 43–55.

Victoria Petroleum 2004, Final well report Wendy-1 stratigraphic drilling project EP 427: Geological Survey of Western Australia, Statutory petroleum exploration report, S20948 A1.

Wingate, MTD, Lu, Y and Allen, HJ 2021, 220109: sandstone, Wendy 1; Geochronology Record 1826: Geological Survey of Western Australia, 9p.

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